

SiteChat Solution Architecture

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Overview

SiteChat gives every CleanTasker Site its own chat room, connecting cleaners, supervisors, managers and clients with real-time messages and automatic system updates (e.g. job completed, inspection failed). This document describes the main components and how they fit together.

Main components

Component	Role
React web app (TypeScript)	Desktop/browser client for supervisors and managers
Flutter mobile app	Mobile client for cleaners and on-site staff
Firebase Authentication	Reuses CleanTasker's existing authentication — no separate login system
Firestore	Stores chat rooms, messages, and site membership (who can access which Site's chat)
Firebase Cloud Functions (Gen 2, TypeScript)	Runs backend logic: posting system messages, enforcing access rules, triggering notifications
Firebase Cloud Messaging (FCM)	Sends push notifications for new messages, mentions, job/inspection updates
Firebase Storage	Stores uploaded photos, videos, PDFs and site documents
CleanTasker REST APIs	Source of truth for Sites, jobs, inspections, clock-in/out and client requests; triggers system messages
Azure-hosted backend	Hosts CleanTasker's existing services that SiteChat integrates with

How it fits together

- A user opens the React or Flutter app and signs in using CleanTasker's existing authentication.
- The app only shows chat rooms for Sites the user is authorised to access (enforced via Firestore security rules).
- Messages, reactions, edits and attachments are written to Firestore and Firebase Storage.
- Cloud Functions listen for CleanTasker events (e.g. job completed) and automatically post a system message into the relevant Site's chat room.
- Cloud Messaging pushes a notification to relevant users' devices for new messages, mentions, or system events.